a. Pixel numbering

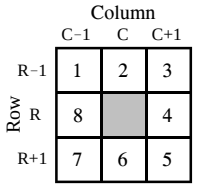
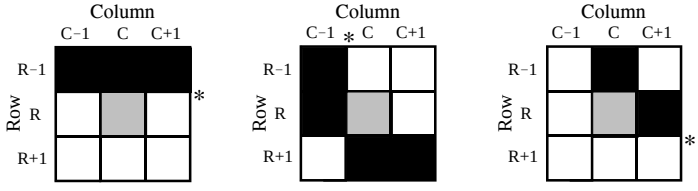
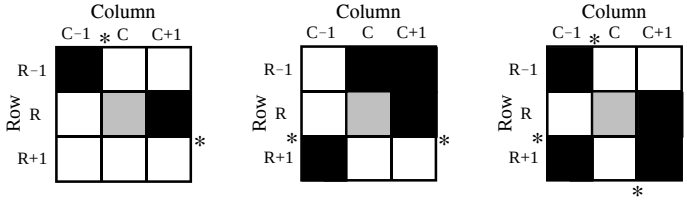


FIGURE 25-12
Neighboring pixels. A pixel at row and column [R,C] has eight neighbors, referred to by the numbers in (a). Figures (b) and (c) show examples where the neighboring pixels are *connected* and *unconnected*, respectively. This definition is used by rule number four of the skeletonization algorithm.

b. Connected neighbors



c. Unconnected neighbors



connected means that all of the black neighbors touch each other. Likewise, *unconnected* means that the black neighbors form two or more groups.

The algorithm for determining if the neighbors are connected or unconnected is based on counting the black-to-white transitions between adjacent neighboring pixels, in a *clockwise* direction. For example, if pixel 1 is black and pixel 2 is white, it is considered a black-to-white transition. Likewise, if pixel 2 is black and both pixel 3 and 4 are white, this is also a black-to-white transition. In total, there are eight locations where a black-to-white transition may occur. To illustrate this definition further, the examples in (b) and (c) have an asterisk placed by each black-to-white transition. The key to this algorithm is that there will be exactly *one* black-to-white transition if the neighbors are *connected*. More than one such transition indicates that the neighbors are *unconnected*.

As additional examples of binary image processing, consider the types of algorithms that might be useful after the fingerprint is skeletonized. A disadvantage of this particular skeletonization algorithm is that it leaves a considerable amount of *fuzz*, short offshoots that stick out from the sides of longer segments. There are several different approaches for eliminating these artifacts. For example, a program might loop through the image removing the pixel at the end of every line. These pixels are identified